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Solving of continuous LTI system responses

- The system has zero input response $y_{zi}(t)$
- The impulse response of the system $h(t)$
- The system responds with zero state $y_{zs}(t)$

Zero input response of a continuous LTI system

※ The definition of the zero input response

※ Solution of zero input response

Definition of the zero input response

The input signal is zero, and the response produced by acting solely on the initial state of the system is called the zero input response, which is denoted $y_{\mathrm{zi}}(t)$

Mathematical models:

$$y^{(n)}(t)+a_{n-1}y^{(n-1)}(t)+\cdots+a_1y'(t)+a_0y(t)=0$$

Solving for zero input responses

The solution process

Step 1: Find the eigengengen for the differential equation;

Step 2: Determine the form of the zero input response based on the feature root;

Step 3: Substitute the initial state into the zero input response representation. Solving for the pending coefficient results in a zero input response.

Solving for zero input responses

The zero input response is in the form of yzi(t).

(1) Feature roots appear unequal solid roots $s_1, s_2, \ldots, s_n$

$$y_{\mathrm{zi}}(t) = K_1 \mathrm{e}^{s_1 t} + K_2 \mathrm{e}^{s_2 t} + \cdots + K_n \mathrm{e}^{s_n t}$$

(2) Feature roots appear equal solid roots $s_1 = s_2 = \ldots = s_n = s$

$$y_{\mathrm{zi}}(t) = K_1 \mathrm{e}^{s t} + K_2 t \mathrm{e}^{s t} + \cdots + K_n t^{n-1} \mathrm{e}^{s t}$$

(3) Characteristic roots appear in pairs of conjugate compound roots $s_i = \sigma_i \pm \mathrm{j}\omega_i$

$$y_{\mathrm{zi}}(t) = \mathrm{e}^{\sigma_i t}(K_1 \cos \omega_i t + K_2 \sin \omega_i t)$$

[Example] Differential equations that describe a continuous LTI system are known $y''(t)+6y'(t)+8y(t)=x(t), \quad t>0$

Initial state $y(0^-)=1$, $y'(0^-)=2$ Evaluate the zero input response $y_{zi}(t)$

Solution: The characteristic equation is $s^2+6s+8=0$

The feature root is $s_1=-2$, $s_2=-4$ (two unequal solid roots)

$$y_{zi}(t)=K_1e^{-2t}+K_2e^{-4t}$$

$y(0^-)=y_{zi}(0^-)=K_1+K_2=1;$

$y'(0^-)=y'_{zi}(0^-)=-2K_1-4K_2=2$

$\Rightarrow$ $K_1=3$, $K_2=-2$

$$y_{zi}(t)=3e^{-2t}-2e^{-4t}, \quad t\geq 0^-$$

[Example] The differential equations that describe a continuous LTI system are known to be: $y''(t)+4y'(t)+4y(t)=3x(t),\ t>0$

The initial state is $y(0^-)=2$, $y'(0^-)=-1$ Evaluate the zero input response $y_{zi}(t)$

Solution: The eigen equation is $s^2+4s+4=0$

The feature root is $s_1=s_2=-2$ (two equal solid roots)

$$y_{zi}(t)=K_1e^{-2t}+K_2te^{-2t}$$

$y(0^-)=y_{zi}(0^-)=K_1=2;$

$y'(0^-)=y'_{zi}(0^-)=-2K_1+K_2=-1$ $\Rightarrow$ $K_1=2,\quad K_2=3$

$$y_{zi}(t)=2e^{-2t}+3te^{-2t},\qquad t\geq 0^-$$

[Example] The differential equations that describe a continuous LTI system are known to be: $y''(t)+2y'(t)+5y(t)=4x(t),\ t>0$

The initial state is $y(0^-)=1$, $y'(0^-)=3$ Evaluate the zero input response $y_{zi}(t)$

Solution: The eigen equation is $s^2+2s+5=0$

The feature root is $s_1=-1+2\mathrm{j}$, $s_2=-1-2\mathrm{j}$ (Two conjugate compound roots)

$$y_{zi}(t)=\mathrm{e}^{-t}(K_1\cos 2t+K_2\sin 2t)$$

$y(0^-)=y_{zi}(0^-)=K_1=1$

$y'(0^-)=y'_{zi}(0^-)=-K_1+2K_2=3$

$\Rightarrow$ $K_1=1,\ K_2=2$

$$y_{zi}(t)=\mathrm{e}^{-t}(\cos 2t+2\sin 2t),\quad t\geq 0^-$$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!